When and Whom to Quarantine? Topology-Aware Two-Tier Control Under Alert-Driven Budget Exhaustion in Agent Swarms

Pre-Sprint Pipeline Validation, Held-Out Replication Audit, and Agentic Zero-Trust Roadmap

Swarm Alignment Hackathon (AIAF × Apart Research) — Pre-Existing Infrastructure Audit

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PRE-KICKOFF COMPLIANCE AND METHODOLOGICAL DISCLAIMER. All numerical results documented in this report were produced exclusively in deterministic `-mode mock` prior to the November 6, 2026 kickoff (zero live LLM swarm calls executed). In accordance with Apart Research rules, these benchmarks validate simulator mechanics, statistical non-inferiority pipelines, Common Random Numbers (CRN) prefix sharing, and figure generation, and **must never be cited as empirical LLM findings** in the final hackathon submission.

Abstract

Multi-agent LLM workflows overseen by imperfect monitors face an acute *alert-driven budget exhaustion* vulnerability: transient false alarms emitted by benign peripheral nodes can deplete a scarce intervention cap (K) before genuine contagion reaches high-centrality hubs. We present the complete pre-sprint engineering validation of `swarmlab`, evaluating 10 control policies across 4 network configurations (`chain`, `small_world`, `star_hub`, `star_leaf`), 2 alert regimes (`clean`, `decoy`), 30 tuning seeds, 30 main seeds ($2{,}400$ episodes), 30 held-out replication seeds ($2{,}400$ episodes), and a targeted scaling sweep ($N \in \{12, 24, 36\}$, 600 episodes). Across both main and held-out splits, all four pre-sprint audit repairs succeed: (1) `fixed_early` and `score_only` follow distinct decision paths and diverge sharply under decoy bursts; (2) benign decoy spikes significantly degrade naive early/score triggers (**H2 supported**: $5/8$ cells on main, $4/8$ on held-out); and (3) our graduated two-tier policy (`risk_aware_tiered`, combining per-step schema sanitization with hard edge isolation) achieves strict Pareto or utility-non-inferior dominance on 68% of comparisons on main ($19/28$) and 64% on held-out ($18/28$) with **zero significant utility losses** ($0/28$), whereas single-tier `risk_aware` reaches 42% ($10/24$). Finally, we formalize the next-stage roadmap framing `swarmlab` as a **Budgeted Agentic Zero-Trust Architecture (BAZT)** with epistemic shrinkage and cryptographic canary tripwires ahead of live LLM execution.

1 Executive Validation Summary and Audit Resolution

An earlier mock audit identified four structural defects: numerical collapse between `fixed_early` and `score_only`, single-objective tuning ignoring collaborative accuracy (MAE), an overly permissive Pareto win label, and uncalibrated decoy pressure. All four defects have been resolved and verified across 45 automated unit tests and $5{,}400$ total validation episodes:

1. **Distinct Operational Semantics and Transient Decoy Realism:** Benign decoy nodes (`[DOC-2]`) now generate realistic 1–2 step early work-cycle bursts ($\mu_{\text{dec}} = 2.8 > \mu_{\text{inf}}/2$) before subsiding, whereas true contagion (`[REF-7]`) persists and spreads. `fixed_early` operates as a snapshot threshold trigger within steps $1 \ldots 4$, whereas `score_only` requires m-step consecutive confirmation across steps $1 \ldots 12$.

2. **Joint Safety–Utility Parameter Selection:** Tuning runs across all 30 dedicated seeds (`9001-9030`) optimizing the composite control objective $J = \text{AUC}_{\text{ever}} + \alpha \cdot (\text{MAE}/\text{MAE}_{\text{base}})$ with $\alpha = 0.25$. On the tuning split, `risk_aware_tiered` selects a soft-first graduated regime ($\tau_{\text{san}} = 0.35$, $\tau_{\text{quar}} = 2.0$, $\lambda = 0.25$).

3. **Corrected Per-Step Tier-1 Mechanics:** Tier-1 `trusted_sanitize()` strips free-text tags on a strictly *per-step* basis ($B_{\text{san}} = 6$ actions/episode, strip probability $p_{\text{strip}} = 0.90$ via shared CRN draws), preserving numeric estimates est $= \langle\text{number}\rangle$ without permanently silencing nodes. Tier-2 candidates exceeding hard cap $K = 3$ gracefully degrade to Tier-1 rather than dropping unprotected.

4. **Unambiguous Statistical Taxonomy:** `stats_check.py` enforces Holm-adjusted paired Wilcoxon criteria separating `STRICT_PARETO_WIN` ($\Delta\text{AUC} < 0, p^{\text{AUC}}_{\text{Holm}} < 0.05$ and $\Delta\text{MAE} < 0, p^{\text{MAE}}_{\text{Holm}} < 0.05$) from `SAFETY_WIN_UTILITY_NONINFERIOR` (upper 95% bootstrap CI of $\Delta\text{MAE} \leq 0.05$).

System Component	Validated Configuration	Audit & Budget Status
Topologies ($N = 12, T = 12$)	`chain`, `small_world`, `star_hub`, `star_leaf`	Distinct hub vs. peripheral initial compromise tested.
Dataset Splits (Disjoint)	Tune: `9001-9030`; Main: `1-30`; Held-out: `31-60`	$2{,}400$ episodes/split; separate CSV and JSONL files.
Control Arms (10 Policies)	`no_defense`, `fixed_{early,mid,late}`, `random_capped`, `score_only`, `centrality_only`, `risk_aware`, `risk_aware_tiered`, `offline_oracle`	Hard quarantine cap $K = 3$; Tier-1 per-step sanitization cap $B_{\text{san}} = 6$ ($p_{\text{strip}} = 0.90$).
Scaling Sweep (`small_world`)	$N \in \{12, 24, 36\}$, seeds `1-20` (600 episodes)	Compares fixed $K = 3$ vs. proportional $K = \text{round}(0.25N)$.
Live LLM Cost Bound	$34{,}560$–$345{,}600$ calls (≈ 89.9M in / 13.8M out max)	**$0.62 – $6.22 USD** at $0.06/Mtok with CRN SQLite cache.

Table 1: Validated experimental matrix and projected live LLM API budget for the 30-seed main split.

2 Grand-Mean Synthesis and Analysis of Figures 1–3

Because external figure binaries are maintained as separate vector artifacts (`results/figures/fig1-fig3.pdf`), Table 2 and Table 3 report the exact numerical coordinates underlying all three diagnostic figures.

2.1 Safety–Utility Frontier (Figure 1) and Budget Exhaustion Dynamics (Figure 2)

In `fig1_pareto_safety_utility.pdf`, plotting containment damage (AUC_{ever}) against consensus estimation error (MAE) across all 10 policies under the decoy regime demonstrates a clear structural separation. `centrality_only` achieves low AUC_{ever} (0.206) by immediately isolating high-degree hubs, but suffers severe consensus error ($\text{MAE} = 11.52$) whenever hubs are healthy. Conversely, `score_only` waits for m-step confirmation, allowing fast hub cascades ($\text{AUC}_{\text{ever}} = 0.367, \text{MAE} = 9.24$). `risk_aware_tiered` occupies the lower-left frontier ($\text{AUC}_{\text{ever}} = 0.169, \text{MAE} = 7.38$), closest to the privileged `offline_oracle` $(0.083, 3.70)$.

Crucially, `fig2_budget_exhaustion_dynamics.pdf` reveals a key diagnostic insight that motivates our v3 pre-sprint upgrade: even `risk_aware` (0.77) and `risk_aware_tiered` (0.67) currently expend roughly two of their three hard quarantines at step $t = 1$ on decoy alerts when a decoy happens to occupy a moderately connected node, because single-step suspicion $p_{i,1}$ is not yet discounted for low sample maturity ($n_{\text{obs}} < m$).

Policy Arm	$\text{AUC}_{\text{ever}} \downarrow$	MAE $\downarrow$	Decoy Waste	Trajectory Profile in Figures 1 & 2
`no_defense`	0.606	19.56	0.00	Unmitigated cascade ceiling; severe bias drift.
`score_only`	0.367	9.24	0.61	Zero spend at $t = 1$; spikes to ~ 1.81 decoy quarantines at $t = 2$–3.
`fixed_early`	0.254	8.97	0.80	Burns ~ 2.40 of $K = 3$ on $t = 1$–2 decoys; only ~ 0.56 hit true positives.
`risk_aware`	0.222	8.89	0.77	Avoids leaf decoys late, but still wastes ~ 2.32 quarantines at $t = 1$.
`centrality_only`	0.206	11.52	0.01	Blind hub isolation; low contagion but worst deployable MAE among top arms.
`risk_aware_tiered`	**0.169**	**7.38**	**0.67**	**Best deployable frontier**: Tier-1 absorbs ambiguity; ~ 2.00 decoy vs. 0.60 TP.
`offline_oracle`	0.083	3.70	0.00	Non-deployable ground-truth upper bound ($1/12$ floor).

Table 2: Main split (`seeds 1-30`) decoy-regime means across all 4 topologies, synthesizing `fig1_pareto_safety_utility.pdf` and `fig2_budget_exhaustion_dynamics.pdf`.

2.2 Swarm Size Scaling Laws (Figure 3: $N \in \{12, 24, 36\}$)

Table 3 documents the exact values plotted in `fig3_scaling_curve.pdf` on `small_world` (20 seeds, 600 episodes). Two clean scaling laws emerge:

- **Scarce Fixed Budget ($K = 3$ as N grows to 36):** When hard interventions remain fixed at $K = 3$ (8.3% of a 36-agent swarm), `risk_aware_tiered` substantially outperforms single-tier `risk_aware` at every scale ($0.267/12.35$ vs. $0.383/18.63$ at $N = 12$; $0.265/13.33$ vs. $0.338/17.33$ at $N = 24$; $0.186/9.56$ vs. $0.221/11.69$ at $N = 36$), while `score_only` degrades to $0.383/18.97$ at $N = 36$.
- **Proportional Budget ($K = \text{round}(0.25N)$):** At $N = 24$ ($K = 6$) and $N = 36$ ($K = 9$), `risk_aware` and `risk_aware_tiered` converge onto near-oracle performance ($\text{AUC}_{\text{ever}} \in [0.033, 0.044]$, MAE $\in [2.95, 3.66]$), demonstrating that soft Tier-1 mediation specifically compensates for *hard-budget scarcity*.

Size N	Budget Rule	No Defense	Score-Only	Centrality	Risk-Aware	Tiered ($R_{i,t}$)
$N = 12$	$K = 3$ ($0.25N$)	0.725 / 42.13	0.428 / 15.73	0.383 / 16.09	0.383 / 18.63	$\mathbf{0.267}$ / $\mathbf{12.35}$
$N = 24$	$K = 3$ (Fixed)	0.580 / 32.37	0.450 / 21.36	0.368 / 18.09	0.338 / 17.33	$\mathbf{0.265}$ / $\mathbf{13.33}$
$N = 24$	$K = 6$ ($0.25N$)	0.580 / 32.37	0.313 / 10.78	0.214 / 10.96	0.042 / 2.96	$\mathbf{0.042}$ / $\mathbf{2.95}$
$N = 36$	$K = 3$ (Fixed)	0.528 / 29.38	0.383 / 18.97	0.366 / 18.84	0.221 / 11.69	$\mathbf{0.186}$ / $\mathbf{9.56}$
$N = 36$	$K = 9$ ($0.25N$)	0.528 / 29.38	0.273 / 8.61	0.175 / 10.20	0.044 / 3.66	$\mathbf{0.033}$ / $\mathbf{3.19}$

Table 3: Targeted scaling sweep on `small_world` (`seeds 1-20`, entries formatted as AUC_{ever} / MAE), corresponding to `fig3_scaling_curve.pdf`.

3 Pre-Registered Statistical Tests: Main vs. Held-Out Replication

3.1 H1: Effect of Intervention Window Timing (Supported on 4/4 Topologies)

Table 4 confirms that under matched cap $K = 3$, intervention window timing significantly shapes cumulative infection across all four topologies on both splits (Early $<$ Mid $<$ Late, all $p_{\text{Holm}} < 0.05$). Consistent with

topological amplification theory [10], early action yields the largest absolute containment gains on hub-dominated and shortcut graphs (`star_hub`: $\Delta\text{AUC}_{\text{E}-\text{L}} = -0.472$ main, -0.399 held-out; `small_world`: -0.429 main, -0.437 held-out) and smaller, more gradual gains on linear `chain` (-0.184 main, -0.122 held-out).

Topology	Window Pair	ΔAUC (Main)	95% CI (Main)	p_{Holm} (Main)	ΔAUC (Hold)	p_{Holm} (Hold)
chain	early - mid	-0.116	$[-0.151, -0.081]$	3.96×10^{-5}	-0.062	0.0254
chain	mid - late	-0.068	$[-0.088, -0.047]$	3.96×10^{-5}	-0.059	1.76×10^{-4}
chain	early - late	-0.184	$[-0.219, -0.147]$	8.51×10^{-6}	-0.122	1.32×10^{-4}
small_world	early - mid	-0.414	$[-0.507, -0.326]$	5.16×10^{-6}	-0.413	5.20×10^{-6}
small_world	mid - late	-0.015	$[-0.025, -0.007]$	9.79×10^{-4}	-0.024	9.73×10^{-4}
small_world	early - late	-0.429	$[-0.520, -0.344]$	5.16×10^{-6}	-0.437	5.20×10^{-6}
star_hub	early - mid	-0.447	$[-0.543, -0.348]$	5.18×10^{-6}	-0.373	5.18×10^{-6}
star_hub	mid - late	-0.025	$[-0.044, -0.009]$	0.00488	-0.026	0.00391
star_hub	early - late	-0.472	$[-0.563, -0.377]$	5.18×10^{-6}	-0.399	5.18×10^{-6}
star_leaf	early - mid	-0.171	$[-0.259, -0.079]$	0.00261	-0.139	0.0355
star_leaf	mid - late	-0.130	$[-0.177, -0.084]$	2.12×10^{-4}	-0.136	1.83×10^{-4}
star_leaf	early - late	-0.302	$[-0.373, -0.226]$	5.75×10^{-6}	-0.275	9.48×10^{-6}

Table 4: H1 paired Wilcoxon signed-rank tests under the decoy regime across main (`seeds 1-30`) and held-out (`seeds 31-60`) splits.

3.2 H2: Alert-Driven Budget Exhaustion (Supported on Both Splits)

Table 5 evaluates $\Delta\text{AUC} = \text{AUC}_{\text{decoy}} - \text{AUC}_{\text{clean}}$ (positive values indicate degraded containment caused by benign decoy chatter). Snapshot early triggers (`fixed_early`) suffer severe, statistically significant budget exhaustion on $3/4$ topologies in main and $4/4$ in held-out (up to $+0.220$ AUC on `star_hub`, $p_{\text{Holm}} = 0.00271$). `score_only` is significantly degraded on both star topologies on main ($+0.053, +0.077$), while exhibiting slight negative point estimates on `small_world` (-0.054 main, -0.064 held-out, $p_{\text{Holm}} > 0.05$) because `score_only` is already heavily delayed on dense graphs and incidental collateral isolation of a peripheral node occasionally severs an infection bridge.

Topology	Policy Arm	ΔAUC (Main)	95% CI (Main)	p_{Holm} (Main)	ΔAUC (Hold)	p_{Holm} (Hold)
chain	score_only	$+0.020$	$[-0.005, +0.044]$	0.124	$+0.033$	0.121
chain	fixed_early	$+0.044$	$[+0.012, +0.084]$	0.0645	$+0.063$	$\mathbf{0.0205}^*$
small_world	score_only	-0.054	$[-0.116, +0.004]$	0.0645	-0.064	0.102
small_world	fixed_early	$+0.162$	$[+0.092, +0.238]$	$\mathbf{0.00855}^*$	$+0.137$	$\mathbf{0.0315}^*$
star_hub	score_only	$+0.053$	$[+0.022, +0.088]$	$\mathbf{0.0195}^*$	$+0.052$	0.102
star_hub	fixed_early	$+0.140$	$[+0.072, +0.222]$	$\mathbf{0.0103}^*$	$+0.220$	$\mathbf{0.00271}^*$
star_leaf	score_only	$+0.077$	$[+0.030, +0.131]$	$\mathbf{0.0256}^*$	$+0.028$	0.278
star_leaf	fixed_early	$+0.147$	$[+0.083, +0.212]$	$\mathbf{0.00643}^*$	$+0.177$	$\mathbf{0.00456}^*$

Table 5: H2 decoy-induced containment degradation ($\text{AUC}_{\text{decoy}} - \text{AUC}_{\text{clean}}$). Asterisks denote Holm-adjusted $p < 0.05$ (`DEGRADED`).

3.3 H3: Graduated Two-Tier Dominance Over Single-Tier Baselines

Table 6 and Table 7 present the complete H3 evaluation. While single-tier `risk_aware` achieves a 0.42 dominance rate ($10/24$, incurring 1 significant MAE loss on `star_hub` vs. `score_only` on main), our two-tier protocol `risk_aware_tiered` passes the pre-registered H3 criterion on **both** splits: 68% dominance

on main (17 strict Pareto + 2 non-inferior wins, 0 utility losses) and 64% on held-out (16 strict Pareto + 2 non-inferior wins, 0 utility losses). Notably, `risk_aware_tiered` strictly Pareto-dominates single-tier `risk_aware` head-to-head on `small_world` ($\Delta\text{AUC} = -0.096, \Delta\text{MAE} = -5.13$) and `star_leaf` ($\Delta\text{AUC} = -0.124, \Delta\text{MAE} = -1.30$) across both splits.

Challenger Arm	Split	Strict Win	Non-Inf.	Cont.-Only	MAE Loss	Dom. Rate	H3 Supported
`risk_aware`	Main (1–30)	6	4	7	1	$0.42\ (10/24)$	False
`risk_aware`	Held-Out (31–60)	7	3	8	0	$0.42\ (10/24)$	False
`risk_aware_tiered`	**Main (1–30)**	**17**	**2**	**3**	**0**	$\mathbf{0.68}\ (19/28)$	**True**
`risk_aware_tiered`	**Held-Out (31–60)**	**16**	**2**	**4**	**0**	$\mathbf{0.64}\ (18/28)$	**True**

Table 6: H3 aggregate dominance scorecard under the decoy regime ($\text{margin}_{\text{MAE}} = 0.05$).

Topology	Rival Arm	ΔAUC_m	$p^{\text{AUC}}_{\text{Holm}}$	ΔMAE_m	$p^{\text{MAE}}_{\text{Holm}}$	Main Label (1–30)	Held-Out Label (31–60)
`chain`	`score_only`	-0.067	0.0563	-0.77	1.00	`NO_SIG_DIFF`	`NO_SIG_DIFF`
`chain`	`centrality_only`	-0.101	$2.04\text{e}{-4}$	-6.82	$3.30\text{e}{-6}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`chain`	`random_capped`	-0.153	$1.13\text{e}{-4}$	-6.29	$1.52\text{e}{-4}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`chain`	`fixed_early`	-0.007	0.879	-1.42	1.00	`NO_SIG_DIFF`	`NO_SIG_DIFF`
`chain`	`fixed_mid`	-0.123	$9.80\text{e}{-5}$	$+0.92$	1.00	`CONTAIN_UNBOUNDED`	`CONTAIN_UNBOUNDED`
`chain`	`fixed_late`	-0.191	$1.21\text{e}{-5}$	-5.63	$1.42\text{e}{-4}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`chain`	`risk_aware`	$+0.036$	0.0358	$+0.55$	1.00	`NO_SIG_DIFF`	`NO_SIG_DIFF`
`small_world`	`score_only`	-0.222	$5.38\text{e}{-4}$	-5.98	0.0291	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`small_world`	`centrality_only`	-0.177	0.00131	-7.32	0.0121	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`small_world`	`random_capped`	-0.394	$1.58\text{e}{-5}$	-16.61	$2.83\text{e}{-6}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`small_world`	`fixed_early`	-0.066	0.0437	-3.39	0.0291	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`small_world`	`fixed_mid`	-0.480	$1.21\text{e}{-5}$	-13.39	$4.99\text{e}{-6}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`small_world`	`fixed_late`	-0.495	$1.21\text{e}{-5}$	-24.23	$1.30\text{e}{-8}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`small_world`	`risk_aware`	-0.096	0.00131	-5.13	$6.80\text{e}{-4}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`star_hub`	`score_only`	-0.337	$1.15\text{e}{-5}$	$+0.62$	0.125	`CONTAIN_UNBOUNDED`	`CONTAIN_UNBOUNDED`
`star_hub`	`centrality_only`	$+0.064$	0.0625	$+0.13$	1.00	`NO_SIG_DIFF`	`NO_SIG_DIFF`
`star_hub`	`random_capped`	-0.581	$1.15\text{e}{-5}$	-1.22	0.0153	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`star_hub`	`fixed_early`	-0.153	$7.32\text{e}{-4}$	-0.37	0.0986	`SAFETY_NONINFERIOR`	`SAFETY_NONINFERIOR`
`star_hub`	`fixed_mid`	-0.600	$1.15\text{e}{-5}$	-0.19	1.00	`CONTAIN_UNBOUNDED`	`CONTAIN_UNBOUNDED`
`star_hub`	`fixed_late`	-0.625	$1.15\text{e}{-5}$	-0.98	0.0986	`SAFETY_NONINFERIOR`	`STRICT_PARETO_WIN`
`star_hub`	`risk_aware`	-0.031	0.0625	-0.18	0.125	`NO_SIG_DIFF`	`NO_SIG_DIFF`
`star_leaf`	`score_only`	-0.165	0.00232	-1.34	0.0496	`STRICT_PARETO_WIN`	`CONTAIN_UNBOUNDED`
`star_leaf`	`centrality_only`	$+0.065$	$9.77\text{e}{-4}$	-2.56	$6.99\text{e}{-6}$	`NO_SIG_DIFF`	`NO_SIG_DIFF`
`star_leaf`	`random_capped`	-0.355	$2.11\text{e}{-5}$	-2.18	$6.63\text{e}{-4}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`star_leaf`	`fixed_early`	-0.116	0.00232	-1.18	0.0071	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`star_leaf`	`fixed_mid`	-0.287	$3.51\text{e}{-4}$	-0.97	0.0496	`STRICT_PARETO_WIN`	`SAFETY_NONINFERIOR`
`star_leaf`	`fixed_late`	-0.417	$1.34\text{e}{-5}$	-1.92	0.00103	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`
`star_leaf`	`risk_aware`	-0.124	0.00146	-1.30	$9.43\text{e}{-4}$	`STRICT_PARETO_WIN`	`STRICT_PARETO_WIN`

Table 7: Complete 28-comparison pairwise evaluation of `risk_aware_tiered` across main and held-out splits.

4 Next-Stage Scientific Roadmap: Budgeted Agentic Zero-Trust (v3)

With the core codebase verified, four high-leverage refinements will maximize scientific novelty for the final 8-page paper while staying 100% within our 48-hour sprint compute envelope:

1. **Formal Agentic Zero-Trust Architecture (ZTA) Framing [6, 7]:** Rather than presenting `risk_aware_tiered` as an ad-hoc heuristic, we formalize it as a *Budgeted Agentic Zero-Trust* controller:

- *Policy Engine (PE):* Continuous per-step trust evaluation $R_{i,t}$ rejecting implicit topological trust (unlike `centrality_only`, hubs enjoy neither immunity nor blind pre-emptive shutdown).
- *Tier-1 Policy Enforcement — Content Disarm and Reconstruction (CDR):* Schema-constrained numeric extraction enforcing least-privilege communication [4, 5].
- *Tier-2 Policy Enforcement — Dynamic Micro-Segmentation:* Permanent outgoing edge isolation once confirmed contagion risk exceeds τ_{quar}.

2. **Epistemic Shrinkage and Neighborhood Corroboration:** To eliminate the early $t = 1$ decoy waste $(0.67–0.77)$ identified in Figure 2, raw suspicion $p_{i,t}$ will be shrunk toward prior p_0 when observed steps $n_{\text{obs}} < m$, with likelihood boosting when upstream neighbors also exhibited suspicion at $t - 1$.

3. **Cryptographic Canary Trap & Byzantine Breakdown Comparison [8]:**
   - *Side-Channel Tripwire:* Assigning each agent i a private watermark `PRIV_KEY_i` provides an exact zero-FPR exfiltration tripwire alongside noisy semantic monitoring (Track 4 integration).
   - *Why Passive Byzantine Consensus Needs Active ZTA:* Adding a lightweight trimmed-mean aggregator (`W-MSR`) demonstrates that passive fault tolerance collapses once contagious compromise exceeds local $1/3$ neighborhood bounds, whereas ZTA containment keeps prevalence strictly inside the resilient regime.

4. **Bridging White-Box Probe Monitors [9, 1]:** Even high-AUROC activation probes with a $2\%–5\%$ per-turn false-positive rate yield $1 - (1 - \text{FPR})^{N \cdot T} > 95\%$ episode-level false-alarm probability across a 12×12 swarm, proving analytically why white-box monitors still require our downstream ZTA budget controller.

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